

SYED ALIYAR SHAH

OVERVIEW OF PROJECT AND RESEARCH PORTFOLIO DOCUMENT

WRITTEN PROJECTS AND RESEARCH/REPORTS

Ohmitron Battery Pack, Chargers and VPMS Provisional Patents and Proposals

- Patented Ohmitron's Li-ion Battery Packs, Chargers, VPMS and converters under US Patents
- Proposed the BPs as **Lead Engineer of Ohmitron and Battery Pack Design Team** to initiate manufacturing and commercialization of BPs.

Design, Analysis and Comparison of NPTs and Hybrid Tires Research Paper:

- Innovation of Hybrid Tire made of Composite Materials and comparison with current Non-Pneumatic Tires

Review Paper on Extreme Conditions Composites and their Applications:

- Detailed review paper on performance of Composite Materials under Extreme Conditions in Nuclear, Automobile, Aerospace and Hydrogen Storage System and Industries

ICRAI Assessment of Impact Resistance Characteristics of 3D-Printed Polymers:

- Investigation of Impact assessments of 3D Printed Polymers including PLA-WM, PLA and ABS based on average damage area of fracture

Trailer Stability System Research Project:

- Trailer Stability System Project proposal to U.S DOT based on Vehicle Dynamics

Tribo-Electric Nanogenerator Powered Electrostatic PV Cell Cleaning System:

- Designed and developed an autonomous, waterless dust removal system for solar **PV panels**.
- Utilized (FEA), mathematical modeling to power **electric field distribution** using **TENGs** and MEMS actuators for **micro-vibration cleaning**.
- Achieved up to **95% dust removal** using **1% PV output** in simulated **desert conditions**.

Battery Materials & Chemistry Video Research

- Produced an **Animated Video** on basis of **Project Report**, of Battery Materials and studied chemistries of **Li-Ion** and **Solid State Batteries**
- Used Video Editing Skills in **Filmora**, published a **Video** based research for **Battery Materials and Chemistries**

Shear Loading in Steel Bolted Beams

- Performed **FEA** to determine critical facet of bridge design by obtaining critical Design Factor of 2 for **4 Bolts Lap Joint assembly** and Theoretical calculations using Machine Design Techniques ([Project Link](#)).

Brinell's Hardness Test on Different Materials

- Performed Brinell's Hardness Test on **Cast Iron**, **Mild Steel** and **High Speed Steel** ([Project link](#)).
- Using **UHT Machine** and **2.5mm dia Carbide ball indenter** and **1839N Load** for **10s**
- Obtained Brinell's Hardness Number (BHN) of **202, 150 & 287** for **Cast Iron, MS & HSS**

CAREER PROJECTS

- Completion of **numerous (50+)** projects in **Academic and Professional career** involving **curricular, extra-curricular, Governmental, National and International Competitions Projects and awards**. Most projects aren't mentioned with some highlighted below:

Design and Development of Glass Fiber Reinforced Composite (GFRP) Insulators for HV Power Transmission:

- Final Year Project** of development of **E-Glass Fiber Reinforced Polymer Matrix Composite (GFRPs with Epoxy Resin Matrix)** and **HTV Silicon Rubber** based Insulators for **36kV** rated Power Transmission Lines and Grid Stations
- Reduces Power Loss and Grounding, Electric Short Circuit Incidents compared to Ceramic Insulators with **200% longer life** using composite materials application
- Fully Sponsored Project** by Composite Material Industry (**Polymer Tek PVT.LMT**)

Battery Pack Frame Fatigue Life Estimation:

- Estimation of Fatigue and Design Life of **Steel based Battery Pack Frame** for Trailer Chassis Mounted Electric Trucks
- Used **Isotropic Materials** Static, Dynamic and Frequency **FE Analyses**

Battery Pack Cell Holder Fatigue/Design life Simulation Analysis:

- Estimation of Battery Pack Cell Holder Fatigue and Design Life made of **3D Printed Polymer (ABS)** and **Composite CFRP Material** for structural integrity.
- Utilized Static, Frequency and Dynamic Loads **FE Simulations**

Ohmitron's proprietary MegaWatt Charging System:

Ohmitron's 9-in-1 Vehicle Power Management System (VPMS)

Ohmitron's Proprietary SiC based Single Phase Power Converters:

SAE BAJA Vehicle Fabrication For Team NUST

- Designed and developed a high-performance **front and rear independent suspension system** for **SAE Baja off-road vehicle** competition.
- Utilized **SolidWorks** for **CAD**, **LOTUS Shark** for suspension geometry optimization according to **camber, caster, toe angles**, and **ANSYS** for structural stress analysis; refined vehicle dynamics through **MATLAB** and **CarSim** simulations and physical testing after manufacturing.
- Delivered a **lightweight**, balanced suspension capable of absorbing high-impact loads while maintaining stability and vehicle control across rugged terrain.

Arduino-Based Battery Management & Monitoring System

- Developed a Battery Management System prototype for **real-time individual cell parameters like Voltages, Current, SOC and Cell Balancing** ([Project Link](#))
- Used **Arduino** for **Cell Monitoring & Management**, **ESP32** for **wireless communication and Monitoring** through **Blynk Server** and **BM3451** for **Cell Balancing**. Successfully achieved project objectives, providing **real-time LCD Display and Online Cell Monitoring accurately** contributing to intelligent energy storage and sustainable solutions.

CFD Analysis of F1 Vehicle, Kenworth T800 Truck & Heat Exchanger

- Performed **Computational Fluid Dynamics** on **F1, Kenworth T800 Truck and Counter Flow Heat Exchangers** using **Solidworks**
- Calculated **Theoretical Values** for Pressure, Velocity, Temperature and **Induced Extruded Profiles** on the Bodies for **Reducing Drag, improving Aerodynamics & Heat Transfer**

Mercedes F1 Vehicle and Steering System Design

- Recreated a **precise and intricate** model of the **Mercedes Formula 1** vehicle for **Dassault Systems** and a Complete Steering System in **Solidworks** and **PTC CREO** as a **Top level Assembly** with **100% real-time systematics**.
- Executed the project in **15 focused on-screen hours over 5 days**
- Dassault Systems Showcased The Vehicle on Media Platforms**